Proof by Induction: Divisibility

Questions 1–17

Question 1. Show that each of the following are divisible by 4.

a) $(2n+1)(2n-3)-4n+23$

b) $16^{3k}-4$

c) $2^{k+2}-4\times 3^k+8$

Question 2. Show that each of the following are divisible by 6.

a) $6^{k+1}-18$

b) $2^{k+1}\times 3^{k+2}$

c) $(k^2+1)(k-4)+10(k^2+k+1)$

Question 3. Given that a, b and c are positive integers, decide whether each of the following statements are true or false:

a) If a and b are factors of c then $a + b$ is a factor of c.
b) If a is a factor of both b and c then a is a factor of $b + c$.
c) If a is a factor of $b + c$ then both b and c are factors of a.
d) If a is a factor of b and b is a factor of c then a is a factor of $b + c$.

Question 4. Prove, by induction, that $n^2 + n$ is divisible by 2 for all integers $n \geq 1$.

Question 5. Using proof by induction, prove that

a) $n^3 - n$ is divisible by 3

b) $n^3 + 3n^2 + 2n$ is divisible by 6 for all integers $n \geq 1$

Question 6. Prove, by induction, that $9^n + 7$ is divisible by 8 for all integers $n \geq 1$.

Question 7. Using proof by induction, prove that $5^n - 4n + 3$ is divisible by 4 for all integers $n \geq 0$.

Question 8. Using proof by induction, prove that $5^n + 4n - 1$ is divisible by 8 for all integers $n \geq 1$.

Question 9. Prove, by induction, that $6^n - 2^n$ is a multiple of 4 for all integers $n \geq 1$.

Question 10. Prove, by induction, that

a) $5^n + 3 \times 9^n$ is divisible by 4

b) $5^{2n} + 3^{2n-1}$ is divisible by 4 for all integers $n \geq 1$

Question 11. Prove, by induction, that $15^n - 8^{n-2}$ is divisible by 7 for all integers $n \geq 2$.

Question 12. Prove, using induction, that for all positive odd integers n, $3^n + 7^n$ is divisible by 10.

Question 13. The following shows a student's answer to the question "Prove, by induction, that $3^{2n}-1$ is a multiple of 8 for all integers $n \geq 1$".

True for $n = 1$
When $n = k + 1$
$(3^{2(k+1)} - 1) - (3^{2k} - 1)$
$= 3^{2k} \times 3^2 - 3^{2k} = 3^{2k}(9 - 1) = 8 \times 3^{2k}$
which is a multiple of 8 so true for all $n \geq 1$

a) Find all the errors.
b) Correct the proof.

Question 14. Prove by induction that the sum of the cubes of three consecutive positive integers is divisible by 3.

Question 15. For $n \geq 1$, it is given that $f(n) = 50n^2 + 70n$.

a) Show that $f(n+1) - f(n) = 20(5n+6)$.

b) Hence prove by induction that $f(n)$ is divisible by 4 for all integers $n \geq 1$.

Question 16. For $n \geq 1$, it is given that $f(n) = 24 \times 2^{4n} + 3^{4n}$.

a) Find $f(n+1) - f(n)$.

b) Hence prove by induction that $f(n)$ is divisible by 5 for all integers $n \geq 1$.

Question 17.

a) Prove that for all integers $n \geq 1$, $f(n) = 2n^3 - 5n$ is a multiple of 3.

b) Show that $f(-x) = -f(x)$ for all real values of x.

c) Hence prove that $f(n)$ is a multiple of 3 for all integer values of n.